PREP 2, LESSONS 3–4

Lifting and stretching

Gravitational potential energy; stretching springs; spring constant

SCORE / 10

Name _____ Class _____ Due _____

Do Now REMEMBER

From earlier lessons, 1 mark each

1 State the standard unit of energy. [1]

2 Name the four ways energy is transferred between stores. [1]

3 Name the two quantities that kinetic energy depends on. [1]

Key ideas LESSONS 3–4

Show your working

4 A 2.0 kg box is lifted onto a shelf 1.5 m high. Use $E_p = m\,g\,h$ and $g = 9.8$ N/kg.
Calculate the energy transferred to the gravitational potential store of the box. [3]

5 A spring is 12.0 cm long. A 3.0 N force stretches it to 18.0 cm. Use $F = k\,e$

a Calculate the extension in metres. [1]

b Calculate the spring constant. [2]

c Up to the limit of proportionality, extension is [1]

directly proportional	inversely proportional	not related

to force.